Mikhail Volkov

AI Researcher / Engineer, PhD in Chemoinformatics

volkowmikhail@gmail.com   Strasbourg, France

https://github.com/mbvolkoff   linkedin.com/in/mikhail-volkov-815743130

Profile

AI researcher / AI engineer with background in data science for life sciences. 6+ years building classical ML and deep neural network based systems in Python / C++ for chemical data; experience with deep learning, experiment tracking, and model packaging. Built agentic AI assistants for data science pipelines. Carried out research projects, co-supervised juniors. Open to domains within and beyond life sciences.

Technical skills

Programming languages
Python (most used), C++, SQL, Shell / Bash

Infrastructure
Linux, Docker, Singularity, slurm (HPC scheduler), git, AWS, FastAPI, CI/CD

UI
Streamlit, Chainlit

Other
Code testing / reporting, collaborative coding, reporting the results of data analysis, writing documentation

ML / DL libraries and frameworks
ML: PyTorch, DGL, PyTorch Lightning, TensorFlow, Keras, scikit-learn, SciPy, ONNX, numpy. Visualization: matplotlib, seaborn, Altair. Optimization: optuna. Data processing: pandas

Agentic AI
Agno, Langchain, Smolagents

ML methods
Classical ML: classification, regression: RF, xgboost, SVM, linear regression. Clustering methods. Deep learning: MLP, GNN, VAE, GAN, CNN, modern LLMs. Finetuning: LoRA

Professional experience

Postdoctoral researcher, 08/2023 – Present
Laboratory of Chemoinformatics - UMR 7140, University of Strasbourg / CNRS Strasbourg, France

Agentic AI system
- Built an agentic data-analysis assistant for chemistry workflows using Agno (backend) + Chainlit (UI), enabling users to upload datasets, run tools, and export results via a chatbot interface.
- Implemented a two-level multi-agent design (planner + specialized agents) and integrated local LLM inference for privacy-sensitive environments.
- Added isolated storage (MinIO S3) and deployed the service with Docker Compose for reproducible installation.

QSAR for cosmetology-relevant molecules
- Designed a combinatorial library of bio-sourced cosmetology-relevant molecules and generated predicted property profiles to guide synthesis prioritization with partner chemists.
- Refactored and optimized a retrosynthesis/enumeration tool, delivering ~100× speedup and enabling rapid iteration over large design spaces.

- Built ML models for multiple endpoints (e.g., skin permeability, antioxidant, antibacterial activity) and performed rigorous dataset curation (duplicates, incompatible assays, error correction). Performed model & parameter optimization - Bayesian (TPE), Genetic algorithm (GA).

Generative AI integration

- Migrated a SMILES generative model from Python/TensorFlow to ONNX & C++ for high-performance, portable inference and static library integration.
- Developed a cross-platform C++ CLI and web UI integrated into the ChemAtlas web service.

PhD Student, *Laboratoire d'innovation thérapeutique – UMR 7200, Faculty of Pharmacy, University of Strasbourg* — 11/2019 – 04/2023, Illkirch, France

Development of graph neural networks for binding affinity prediction in protein-ligand complexes

- Designed and trained a graph neural network for protein–ligand binding affinity prediction using PyTorch & DGL, with reproducible training via PyTorch Lightning.
- Ran feature engineering and ablation studies (node/edge features, pooling choices) and built classical ML baselines for fair comparison.
- Achieved +0.20 Pearson r and −0.16 RMSE compared with a strong CNN baseline on the same benchmark -- PDBBind 2019 hold-out set.
- Performed data augmentation with docking poses, increasing the training set size by 1–2 orders of magnitude
- Deployed training on HPC using Docker / Singularity; implemented SQL-backed batch sampling to stream training data efficiently.

Graduate thesis, *Laboratory of medicinal chemistry, Faculty of Chemistry, Lomonosov Moscow State University* — 2017 – 2019, Moscow, Russia

Blind docking for detection of protein binding sites of bioactive small molecules

- Developed an unsupervised clustering pipeline of docking results for binding-site detection in symmetric proteins, enabling consistent pocket grouping across symmetric chains.
- Implemented data processing and clustering in Python (pandas, scikit-learn).
- Built interactive 3D visualization pipeline with Python & NGLView.

Education

PhD in Chemistry and Chemoinformatics, — 2019 – 2023
École doctorale des Sciences Chimiques - ED222, University of Strasbourg — Strasbourg, France

Specialist degree (Bachelor + Master) in Fundamental and Applied Chemistry, — 2013 – 2019
Department of Chemistry, Lomonosov Moscow State University — Moscow, Russia

Languages

English
C1, IELTS (7.5) - 2019

French
B2, TCF IRN - 2025

German
B2, DSH - 2016

Soft skills

Communication skills
Scientific communication: public talks, scientific writing

Management skills
Co-supervised a master and a PhD student. Cooperated with academic and industrial partners.

Publications

On the Frustration to Predict Binding Affinities from Protein-Ligand Structures with Deep Neural Networks ↗, *Journal of Medicinal Chemistry* — 24/05/2022

Development and benchmarking of modular message-passing GNNs for structure based protein-ligand affinity prediction. It was demonstrated that training and benchmarking results on PDBbind are strongly driven by protein/ligand dataset bias, limiting model generalization to new targets and chemotypes

Spherical GTM: A New Proposition for Visualization of Chemical Data ↗, *Molecular Informatics* — 16/06/2025

Implementation of the Generative Topographic Mapping algorithm for the spherical manifold, several case studies of application of this method to problems relevant for chemoinformatics

ChemSpace Copilot: Agentic AI for Interactive Visualization and Exploration of Chemical Space ↗, *Preprint* — 02/03/2026

Introduction of an agentic AI system for molecular cartography assisted data analysis in chemistry

Participation in conferences

Applicability of graph neural networks to binding affinities prediction from protein-ligand structures, *17th German Conference on Cheminformatics* — 05/2022, Garmisch-Partenkirchen, Germany

- Poster presentation

Other interests

Music

Membership in an orchestra of Breton music.

Dancing

Membership in a Breton folk dancing club